The effect of crude oil price shocks on macroeconomic stability in Ghana

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Abstract

The paper studies the effects of underlying shocks of crude oil price movements on the stability of macroeconomic aggregates in Ghana. We develop a structural vector autoregressive to disentangle the sources that have driven shocks in crude oil market and estimate the effects of the identified shocks on macroeconomic aggregates and on three bilateral exchange rates in Ghana. In addition, we investigate the extent of transmission of the identified shocks on food and non-food price levels. The findings show that shocks by oil supply and demand-specific have a significant effect on real GDP. Further, the identified shocks have a weighty effect on the Ghana/Euro bilateral exchange rate. The findings also suggest that all the identified structural shocks have significant effects on food and non-food inflation. This implies that oil market shocks are detrimental to food and non-food inflation in Ghana.

1. Introduction

Since the 1970s, economists have been interested in the empirical evidence that shows that macroeconomic performance of a country may have some relationships with oil price shocks. This was motivated by the presence of an increasing reliance on oil imports, unparalleled interruptions in the international oil market and weak macroeconomic stability in many countries of the world. It was therefore obvious for economic agents to suspect oil price shocks may be responsible for fluctuations in macroeconomic aggregates. As a result, a staggering amount of empirical studies supported by theory has been undertaken to study the effect of oil price shocks on macroeconomic aggregates. These studies indicate that although the ultimate impact of oil price shocks on macroeconomic aggregates is of similar magnitudes, the channels of transmission are completely different among countries due to the structure of their economies, energy intensity and as to whether they are importing or exporting countries.

In 2007, Ghana found huge deposits of crude oil, and by the close of the fourth quarter of 2010, the country had become the much-anticipated oil and gas economy. Projections of oil production from the Jubilee Fields[1], as well as the exploration of other hydrocarbon sites, indicate that Ghana is likely to become a net exporter of crude oil in the future. The total volume of crude oil production in 2011 was about 24 million barrels, and this rose to a total of over 28 million barrels in the fourth quarter of 2016 (Ministry of Finance, 2017). The cumulative production of crude oil in Ghana from 2011 to 2016 amounted to over 198 million barrels. The associated revenue from the sale of oil rose from about $181 million in 2011 to over $2 billion in the second quarter of 2017 (Ministry of Finance, 2017).

The direction and extent to which crude oil revenue will influence macroeconomic aggregates are contingent on the price, the volumes of exports and the management of the revenue. Before the export of crude oil in commercial quantities in Ghana, the price had reached a record high of $147 per barrel by the middle of 2008. Although projections indicate that the international price of oil will moderate in recent times, they are anticipated to remain more than 75 per cent higher than in 1999. The increases in oil prices generate considerable amount of wealth to the exporting countries such as Ghana. Alternatively, increases in the price of oil possess the potential of causing recession, inflation, appreciation of the domestic currency, low productive activity and low growth of importing countries.

The high oil prices experienced between 1999 and 2008 were driven by many factors including the staggering economic growth of China and India as well as disruption of supply from the producing countries. However, the reaction of macroeconomic aggregates to oil price hikes will differ based on the sources of shocks. Unfavourable oil supply shocks due to production disruption usually cause low oil production, high production cost fuelling inflation and dampening economic activity. To this end, monetary authorities of oil-importing countries need to make a trade-off between inflation stabilisation and output. Similarly, oil price hikes could also be attributed to high demand for oil, brought by high economic growth or precautionary demand. Oil-exporting countries will accommodate the increase in demand by producing more oil and increasing investment in exploration of new hydrocarbon sites. On the other hand, the monetary authorities of oil-importing countries will experience rising inflation but the change in output will be different (Peersman and Van Robays, 2009). This is possible when part of the wealth generated from the export of oil is obtained through increased trade, which reduces the adverse impact on output. On this basis, macroeconomic aggregates of oil-importing countries will not be affected the same way since their Central Banks will not necessarily have to make a balance between price stabilisation and output.

There is evidence of scanty works on the macroeconomic effect of oil price and the precise channels of transmission which makes it difficult to decide on suitable monetary policy reaction in Ghana. A research of this nature is very important since Ghana is both an exporter and importer of oil. The country exports the crude but imports the refined oil for domestic consumption. For instance, between 2010 and 2016, Ghana exported 161,373,478 but imported 3,451,703 barrels of oil (Energy Commission, 2016). Even though total oil imported within this period constituted about 2.0 per cent, in terms of value, it is relatively higher since the refined oil is comparatively more expensive. In this regard, it is unlikely that Ghana will be shielded from any source of oil price shocks.

Moreover, the extant studies demonstrate that the extent and duration of the effect of oil price shocks on the macroeconomy have moderated since the 1990s. Herrera and Pesavento (2009) attributed this to the variations in the structure of an economy and the introduction of systematic monetary policy. In 2002, Bank of Ghana in an attempt to reduce the harmful impact of oil price disturbances and related exogenous shocks on the economy adopted inflation targeting to ensure price stability. On this basis, the price stability is measured by the inflationary target of government and it is revised annually and stated in the budget statement of each fiscal year. The important question is: Has the Ghanaian economy experienced relative stability in food and non-inflation and improved macroeconomic performance with the existence of oil price disturbances over the years? Even though this question is very important, there is, however, very little empirical evidence to show whether the stabilisation strategies initiated over the years have impacted positively or otherwise in keeping inflation stable. On this note, we close these research voids by evaluating the effects of oil price shocks and track the stability of food and non-inflation since the adoption of inflation targeting framework in Ghana.

Even though fiscal and technological shocks are equally important, we concentrate on oil price disturbances for the following explanations. First, shocks in oil price are believed to have caused most of the recessions in the world since the 1970s (Hamilton, 1983). Second, years of oil price shocks are relatively easy to detect compare to other structural shocks. Furthermore, oil price increases have generally being problematic to policymakers looking forward to reduce trade-off between increasing price level and output. Since previous studies have emphasised the negative effect of oil price shocks on macroeconomic aggregates as well as the need to model policy to respond to moderate the effect of oil price shocks on the economy, we pay attention to the sources of oil price shocks and its effect on macroeconomic aggregate and track the stability or otherwise of food and non-inflation since the introduction of inflation targeting framework in moderating the harmful effects of oil price shocks.

Pursuant to the foregoing, we seek to address the effect of oil price shocks on the stability of macroeconomic aggregates in Ghana. Applying structural vector autoregression, we first ask whether oil price shocks are attributed to supply and demand-side activities. In addition, we attempt to evaluate the pass-through of the structural disturbances on food and non-inflation. Finally, we examine whether bilateral exchange rates in Ghana reactin different ways to the recognised oil market shocks. These issues are of significant importance to both the academic community and policymakers.

We show that oil supply and oil-specific demand shocks possess weighty impact on real GDP. Also, the identified shocks have substantial impact on the Ghs/Euro bilateral exchange rate. We further find evidence of structural shocks on food and non-food price levels which indicates oil market shocks are detrimental to food and non-food inflation in Ghana. The rest of the paper is structured as follows: Section 2 reviews the relevant past works related to the study. Section 3 presents the data and the empirical approach. Section 4 discusses the empirical results, while Section 5 concludes the paper with policy recommendations.

2. Review of related literature

There exists a huge amount of studies on oil price shocks and the macroeconomy for both developing and advanced countries. More importantly, the extant literature covers various strands on the impact of oil market disturbances on the stability of macroeconomic aggregates. Earlier studies have concluded that disturbances in oil market are important drivers of recessions through sectoral shifts in economic activity (Hamilton, 1983, 2003). In these studies, Hamilton observed that oil is an intermediate good, which has the potential of reducing the marginal efficiency of capital which can depress future investments leading to low economic activity. In addition, oil price shocks reduce effective demand for oil-driven goods such as automobiles. This shift in demand has the potential to cause a reallocation of inputs particularly labour across sectors of the economy. If the reallocation of labour is expensive, then there is the likelihood of huge reduction in the addition of value across sectors. According to Bernanke (1983), the uncertainty about future profitability of investment induces firms to delay investing in a way to investigate whether the oil price shocks are temporary or persistent which reduce capital accumulation. However, Rotemberg and Woodford (1996) and Barsky and Kilian (2004) argued that these channels do not have the potency to generate the quantitative effect ascribed to oil market disturbances. Rather, the authors attributed adverse effect of oil price disturbances on sectoral productivity to capital obsolesce and wage rigidity. Capital obsolesce makes a large portion of capital obsolete prematurely and wage rigidity increases unemployment of labour attributable to oil price shocks, which together depress productivity.

Similarly, the literature examined whether oil market shocks have a causal impact on economic growth and other macroeconomic aggregates (Cunado and De Gracia, 2005; Chen and Chen, 2007; Du *et al.*, 2010; Wang, 2013). These studies recognised oil as an important predictor of economic performance and modelled the connection between oil and economic performance using production-based framework. Their evidence reveals that rising oil price depresses economic performance because high oil price generates an externality which impedes growth. With regard to macroeconomic aggregates such as consumption, the intuition is that an unfavourable oil supply shock is expected to push oil price high. Following this, oil prices are unlikely to be mean reverting, and therefore, expected shocks will persist. The persistence of these shocks commands huge effect on wealth and decreases the marginal utility of the consumer.

In addition, the extant literature have examined the nature and linkage between monetary policy and crude oil price shocks particularly for advanced economies (Bernanke *et al*., 1997, 2004; Leduc and Sill, 2004; Peersman and Van Robays, 2009). Using the inflationary channel, the studies revealed that the adverse impact of oil price disturbances on production was significant and strong whenever central banks tighten monetary policy as a reaction to the shocks. This implies monetary contraction to control inflationary effects explains practically all the adverse impact of oil price shocks on output. Peersman and Van Robays (2009), however, revealed that the response of monetary authorities of European Central Banks and the Federal Reserve to oil market shocks is markedly different. Monetary authorities of the Euro area appear to attach much importance to inflation stabilisation objective compare to that output than the Federal Reserve.

Finally, the literature have estimated the asymmetric effect of crude oil price shocks. The evidence suggested that macroeconomic aggregates reacted asymmetrically to both negative and positive oil market shocks (Mork, 1989; Balke *et al*., 2002; Peersman and Van Robays, 2009; Atems *et al*., 2015; Nusair, 2016). The implication is that oil market shocks drive significant greater impacts on macroeconomic aggregates than stable oil prices. Generally, most of these literature attribute the channels of the asymmetric responses to adjustment in costs, financial stress and monetary policy.

Most of the studies reviewed concentrate on advanced countries, net oil producing and consuming economies to the neglect of developing countries particularly, those that are currently exporting quite a substantial amount of crude oil. More importantly, these studies capture the oil price shocks applying the Cholesky factorisation techniques in which oil prices are permitted to impact macroeconomic aggregates contemporaneously but where the reverse possesses no significant impact on oil price. Similarly, Jumah and Pastuszyn (2007) studied oil price fluctuations on monetary policy and total demand for Ghana by placing strict exogeneity on oil price without exploring the fundamental sources of shocks. The identification assumptions applied by previous studies particularly Jumah and Pastuszyn (2007) will generate biased results if developments in economic activity of Ghana have influence on the global price of oil contemporaneously. In addition, Jumah and Pastuszyn (2007) applied the vector autoregressive (VAR) as their identification strategy and used impulse response functions as tools to estimate the connection between the variables. However, VARs are reduced-form models which require structural limitations to estimate all important innovations and impulse responses (Lutkepohl, 2005). Without these identified structural restrictions, the impulse responses possess no meaningful economic interpretations.

With regard to Jumah and Pastuszyn (2007), this study differs in many respects. First, this study employs an empirical technique that is built on structural VAR framework using sign limitations to estimate the outcomes of oil price shocks on the stability (or instability) of macroeconomic aggregates in Ghana. This identification technique is applied to differentiate between oil price shocks attributed to exogenous disturbances in the production of oil, world economic activity and oil-specific demand and evaluate the estimated shocks on real GDP, interest rates, price level, real effective exchange rate and money supply. Second, the study includes Ghana's bilateral nominal rates with the United States, Britain and the Euro area to investigate the exchange rate channel through which oil market shocks affect macroeconomic variables indirectly in Ghana which is absent in most studies. This permits for contemporaneous interactions between macroeconomic aggregates between Ghana and its major trading partners. Finally, the study disaggregates inflation into food and non-food inflation and examines their responses to the structural shocks.

3. Methodology and data

To estimate the effect of crude oil price shocks on macroeconomic stability, we employ the structural vector autoregressive (SVAR) technique following a two-step approach. In the first stage, we apply the SVAR to investigate the sources of shocks. Specifically, we use the underlying assumptions associated with SVAR to extract the structural shocks such as oil supply shocks, aggregate demand shocks and market-specific demand shocks that affect oil price. Accordingly, supply shocks affect present availability of crude oil attributed to both positive and adverse supply-side factors. Similarly, demand-side shocks influence the present available quantity of crude oil due to fluctuations in world business cycle. Finally, the market-specific shocks[2] are attributed to the precautionary demand for crude oil due to anticipated fall in the supply over the demand.

In the second stage, we explore the impact of the estimated structural shocks in the first stage on macroeconomic aggregate including real growth of national income, inflation, interest rate, exchange rate and money supply. According to Lorusso and Pieroni (2018), the two-stage approach offers two advantages. First, it allows for relatively easy management of the amount of variables in the SVAR which reduces the estimated conditions connected with assessing a bigger model. Second, decomposing the structural shocks in the crude oil price prevents the use of further identification restrictions on the macroeconomics variables.

3.1. The Structural VAR model

We examine the dynamic effect of oil price shocks by adopting a reduced-form SVAR with a dynamic representation proposed by Sims (1980). Following Kilian (2009), the study constructed an SVAR model by utilising monthly data for $y_t = (\Delta oilprod_t, rop_t, rea_t)'$ where $\Delta oilprod_t$ denotes the percentage variation in the production of global oil, rop_t represents the real price of global oil, and rea_t is world economic activity index developed by Kilian (2009). The rop_t and rea_t variables are transformed using logarithm. The SVAR is therefore expressed as:

$$A_0 y_t = \alpha_t + \sum_{l=1}^{24} \beta_l y_{t-l} + u_t \tag{1}$$

where u_t represents the vector of the reduced-form residuals which are serially and mutually unrelated structural innovations, α_t defines a vector of constants, and A_0 is a lower triangular matrix. Following Lutkepohl (2005), the study assumes that A_0^{-1} possesses a recursive structure which allows the reduced-form error term ε_t to be disintegrated into $\varepsilon_t = A_0^{-1} u_t$.

Since this study does not focus on testing any particular model, the identification restrictions depend on economic theory and intuitions. To this end, we apply short-run restrictions on the simultaneous relations since the SVAR produces efficient results when fitted to short-run restrictions (Basnet and Upadhyaya, 2015). As a way to retrieve the structural shocks of oil price ε_t, associated with the reduced-form innovation u_t, we apply Cholesky factorisation as the identification strategy and express it as:

$$\varepsilon_t = \begin{pmatrix} \varepsilon_t^{\Delta oilprod} \\ \varepsilon_t^{rop} \\ \varepsilon_t^{rea} \end{pmatrix} \begin{pmatrix} a_{11} & 0 & 0 \\ a_{21} & a_{22} & 0 \\ a_{31} & a_{32} & a_{33} \end{pmatrix} \begin{pmatrix} u_t^{oilsupplyshocks} \\ u_t^{oildemandshocks} \\ u_t^{market-specificshocks} \end{pmatrix} \tag{2}$$

According to Kilian (2009), the assumptions which can be identified to underline equation (2) infer that first, oil supply does not react to shocks originating from total demand or oil-specific shocks simultaneously within the same month. This is reflective of the notion that there exist huge costs with altering production in the month which has the potential of demotivating oil producers from changing output instantaneously when aggregate demand changes due to instabilities in the underlying demand source (Atems *et al.*, 2015). Therefore, oil supply shocks are instabilities which move the supply curve that are not driven by macroeconomic fundamentals, but rather can be attributed to overproduction due to poor forecasting, conflicts and quotas imposed by the producing economies. Second, aggregate demand responds to oil supply shock within a month but unresponsive to an oil-specific shock. This also reflects the view that the aftermath of an adverse shock usually leads to a plummeting global production and an increase in oil prices which prevents the expansion in global economic activities. This implies that all shocks leading to oil production and oil prices comoving are associated with demand-side shocks which shift the demand curve. Therefore, aggregate demand is counter-cyclical to oil supply shocks. Third, oil market-specific shock responds to total demand and oil supply shock within a month. This is decomposed into two scenarios: (i) future expectations of a hike in oil demand could influence precautionary buying, speculation and hoarding resulting in increased fluctuations in oil price; (ii) improvements in macroeconomic stability underlie some disturbances in demand for oil as a factor of production, leading to a reduction in volatility in oil supply.

An examination of these three hypotheses requires a clear distinction between shocks emanating from supply-side, global economic activities and those attributed to market-specific demand. For instance, a decoupling of oil-specific demand shock from global demand shock enables the study to impose the constraint that positive global demand disturbances explicitly raise economic activities of the world on the one hand and market-specific demand shock not driven by macroeconomic fundamentals do not influence world economic activities on the other hand. Therefore, the sign restrictions on the market of crude oil are adequate in capturing the three decoupled categories of oil shocks.

Following Peersman and Van Robays (2009) and Baumeister *et al*. (2010), the study performs the Cholesky ordering in equation (2) and imposes theoretical sign restrictions on each computed impulses responses to retrieve the fundamental structural oil shocks. We summarise the identification restrictions in **Table 1**.

We evaluate the identification by investigating whether historically recognised oil shocks influence the performance of the market. We address this by plotting the historical movements of the recognised oil market shocks. In addition, the study judges the identification of the assumptions by investigating whether key macroeconomic variables of Ghana such as real GDP, CPI, interest rate, exchange rate and money supply react in the anticipated manner. However, there are no monthly data on real GDP[3] in Ghana which make it difficult to estimate the reaction of real GDP to the recognised

Table 1 Identification restrictions

	$\Delta oilprod$	rop	rea
Oil supply shock	≤ 0	≥ 0	≤ 0
Oil-specific demand shock	≥ 0	≥ 0	≤ 0
Global demand shock	≥ 0	≥ 0	>0

Source: Peersman and Van Robays (2009) and Baumeister *et al*. (2010).

structural shocks since the shocks are generated applying monthly data. A frequently used solution is to build an equivalent SVAR model (see equation (1)) by aggregating the data to quarterly frequency. But this implies that application of quarterly data will cause the identifying assumptions to lose their credibility (Kilian, 2009; Smiech et al., 2015). Rather than following Kilian (2009) to generate estimates of quarterly shocks by transforming the monthly structural innovations into quarterly innovations, this study used the Bank of Ghana real composite index of economic activity (CIEAReal) compiled in monthly frequency since 2000 to proxy monthly real GDP. This is justified on the grounds that the CIEAReal tracks the GDP closely and measures business confidence in Ghana. Given the absence of feedback from the macroeconomic variables in each month to the structural shocks, the study treats the shocks as predetermined.[4] We estimate the 24-monthly cumulative impulse response functions for the macroeconomic aggregates and their 95 per cent confidence intervals based on the following regressions:

$$\Delta y_t = \theta_j + \sum_{i=1}^{24} \delta_{ij}\hat{\gamma}_{jt-i} + \omega_{jt} \tag{3}$$

$$\pi_t = \phi_j + \sum_{i=1}^{24} \alpha_{ij}\hat{\gamma}_{jt-i} + \varepsilon_{jt} \tag{4}$$

$$\Delta r_t = \beta_j + \sum_{i=1}^{24} \sigma_{ij}\hat{\gamma}_{jt-i} + \varepsilon_{jt} \tag{5}$$

$$\Delta m_t = \vartheta_j + \sum_{i=1}^{24} \kappa_{ij}\hat{\gamma}_{jt-i} + v_{jt} \tag{6}$$

$$reer_t = \varphi_j + \sum_{i=1}^{24} \zeta_{ij}\hat{\gamma}_{jt-i} + u_{jt} \tag{7}$$

where $j = 1,2,\ldots,24$, Δy_t denotes percentage change in real composite index of economic activity, π_t[5] represents inflation, Δr_t denotes nominal interest rate, Δm_t is money supply, $reer_t$ represents the real effective exchange rate,[6] γ_{jt-i} captures the estimate of monthly shocks computed in equation (2), and $j^{(-th)}$ denotes the structural shock in the i^{th} month. The impulse response coefficients in equations (3) to (7) are captured by $\delta_{ij}, \alpha_{ij}, \sigma_{ij}, \kappa_{ij}$ and ζ_{ij}, respectively. Generally, lag selection criteria are not relevant because the highest length of the impulse response function is fixed at 24 months in equations (3) to (7). In addition, there is the possibility of serial correlation among $\omega_{jt}, \varepsilon_{jt}, \varepsilon_{jt}, v_{jt}$ and u_{jt} in equations (3) to (7) as inference on the response estimates is conducted. This can be addressed by calculating the standards errors for the impulse response functions through the application of the fixed-design wild bootstrap technique suggested by Goncalves and Kilian (2004).

The empirical specifications in equations (3) to (7) assume the absence of feedback from $\delta_{ij}, \alpha_{ij}, \sigma_{ij}, \kappa_{ij}$ and ζ_{ij} within a month to γ_{it-i} for $ij = 1, 2, 3, \ldots, 24$ contemporaneously renders the identification correct. This can be attributed to either the Ghanaian economy is small or due to the lack of comovement of economic developments in Ghana and with the global economy.

3.2. Variance decomposition

Finally, we evaluate the precise pass-through of oil price shocks on inflation (food and non-food) and deduce whether the pass-through is driven by direct or indirect effects applying the variance decomposition technique. Even though the ultimate impact of oil price disturbance on macroeconomic indicators are of similar magnitudes, the channels of transmission are completely different. For instance, inflation in Ghana can directly increase oil prices in the consumer basket or indirectly through the rising cost of production on non-oil goods and service. The speed of pass-through to the consumer also can be detected depending on the oil intensity, introduction of regulatory changes of oil market and variations in the configuration of vehicles usage in Ghana.

3.3. Asymmetric effect

The discussions so far focused on the direct connection between oil shocks and macroeconomic variables. This can imply that if a rise in oil price causes a recession, then oil price decreases should drive increased economy activity through the same pathways operating in the reverse direction. However, a good number of previous studies have established asymmetric relationship particularly between output and global oil shocks. Mork (1989) and Hamilton (2003) revealed that oil price increases are significant drivers of GDP growth than oil price decreases. In effect, oil price shocks that are positive and large matter, but the negative shocks are not important. Lee *et al.* (1995) also showed evidence that stable oil prices are important drivers of economic activity than oil price changes in an unstable environment. In Ghana, there is a high likelihood of the presence of asymmetric connection between oil price shocks and macroeconomic aggregates because of the following reasons. First, given that Ghana is both an importer and exporter of crude oil, both large favourable and adverse oil shocks may exert different impacts on the macroeconomy. Second, exogenous oil price increases reflect significantly in domestic petroleum prices both in magnitude and in timing than a fall in world crude oil price. Finally, different economic agents respond in different ways to dissimilar oil shocks. Therefore, we test the hypothesis that macroeconomic aggregates of Ghana response differently to oil market shocks based on whether they are big and positive or negative.

In evaluating the asymmetric effect of oil shocks on macroeconomic aggregates, we follow Hamilton (2003) and Mork (1989) statistical approaches but permit the inclusion of positive and adverse oil price shocks. Explicitly, we model the growth rate of the macroeconomic aggregates (Δma_{it}) as an AR(12) process as:

$$\Delta ma_t = \alpha + \sum_{j=1}^{12} \beta_j \Delta ma_{t-j} + \sum_{j=1}^{12} \lambda_j \Delta x_{t-j}^{+} + \sum_{j=1}^{12} \theta_j \Delta x_{t-j}^{-} + \varepsilon_t \tag{8}$$

where $\varepsilon_t : \left(0, \sigma_i^2\right), x_{t-j}^{+}$ and x_{t-j}^{-} denote oil price shocks using the superscripts as the directions of impact. Equation (8) permits the estimation of asymmetric reactions of the macroeconomic variables to oil shocks through the coefficients λ_j and θ_j.

Hamilton (1983) defines oil price shocks as the logarithmic variation in the oil price with an implied postulation that the impact of oil price shocks has symmetric effect on economic growth, that is $\sum \lambda_j = \sum \theta_j$. In this case, little shocks in oil price exhibit favourable effects on economic activity that is similar to larger shocks. This assumption may be caused by the unwillingness of economic agents to modify their behaviour with the occurrence of little oil price shocks. To this end, Hamilton (1996) indicates that changes in oil price will exert greater effect on economic activity when the current price increases more than the maximum in the previous year. In this case, if the price of oil in the current month exceeds the maximum value of the previous year, we estimate the percentage variation of the maximum value of the previous year, and the series is defined as one. Similarly, if the price of oil in the present month is lower than the value of the previous month in the previous year, the series is captured as zero for that period (Engemann *et al.*, 2014). Thus, in line with Hamilton (1996), a positive oil price shock can be defined as:

$$\Delta x_t^{+} = \max\left\{0, 100 \times \ln \frac{x_t}{\max\{x_{t-1} \ldots x_{t-12}\}}\right\}$$

We based this definition on the assumption that only positive shocks drive economic performance. On the other hand, shocks that are negative can be expressed as:

$$\Delta x_t^{-} = \min\left\{0, 100 \times \ln \frac{x_t}{\min\{x_{t-1} \ldots x_{t-12}\}}\right\}$$

It is important to note that studies such as Blanchard and Gali (2007) and Blanchard and Riggi (2013) have recorded significant time variant between oil shocks and macroeconomic aggregates. To capture this, the study computes a single structural break applying a conventional sup-Wald test. Mork (1989) attempted to relax this assumption in the estimation of potential asymmetry but considered the logarithmic variations in the oil price as the initial shock. Specifically, we capture the asymmetric response to the shocks by modelling oil price increases and decreases as separate variables as follows:

$$\Delta x_t^+ = \begin{pmatrix} x_t & \text{if } x_t > 0 \\ 0 & \text{otherwise} \end{pmatrix}$$

$$\Delta x_t^- = \begin{pmatrix} x_t & \text{if } x_t < 0 \\ 0 & \text{otherwise} \end{pmatrix}$$

3.4. Data

We use monthly data spanning January 2000 to February 2016. In estimating the SVAR model, we employ the percentage variation in total world production of oil $(\Delta oilprod_t)$. We estimate using the log difference of global oil production calculated in millions per barrels produced in a day. These data were gleaned from the US Department of Energy. The index of world economic activity rea_t is estimated by applying the constituents of global real economic activity which determine demand for manufactured goods in the world markets Kilian (2009). The index is determined using a single dry cargo voyage ocean freight rate. This implies high freight rates are strong indications of high global demand for commodities.[7] Real oil price rop_t measures the spot price of Brent crude oil. We deflate the nominal oil price using Ghana's CPI to obtain the real oil price. $CIEAReal_t$ represents the real composite index of economic activity compiled by the Bank of Ghana in monthly frequency since 2000 to proxy monthly real GDP. We estimate this series through the log difference of real composite index of economic activity seasonally adjusted ($US). The interest rate variable is measured by the short-term (3 months) nominal treasury bill rates also sourced from the Bank of Ghana. We sourced real effective exchange rate $(reer)$ and the data from the International Financial Statistics (IFS) dataset portal. The bilateral exchange rates included Ghana cedi/U.S. (Ghs/$US), Ghana cedi/Euro (Ghs/Euro) and Ghana cedi/Great Britain Pounds Sterling (Ghs/GBP). The exchange rate is defined as the price of the Ghana cedi in relation to the foreign currencies so that a rise in it signifies an appreciation while a decrease corresponds to a depreciation of the exchange rate. Bilateral nominal exchange rate data are sourced from the Pacific Exchange Rate Service database.

4. Results

To ensure that the adopted VAR model is free from spurious problems as outlined by Granger and Newbold (1974), we commence the analysis by pretesting the stationarity features of the data. The augmented Dickey–Fuller unit root test technique was applied to all the variables. **Table B1** (see appendix) shows the stationarity test results. The findings report the series display unit root at their levels but became stationary at first difference. In addition, a visual inspection of the plot of twelve seasonal differences of the variables shows stationarity in **Figure A1**, which confirms the test results obtained from the augmented Dickey–Fuller unit root tests. To this end, the estimated VAR model was based on an I(0) process.

4.1. Decomposing shocks in oil market

To observe the effect of oil price shocks on macroeconomic stability, we plot the historical development of the recognised sources of oil shocks in **Fig. 1** emanating from equation (2). In judging the identification, we evaluate whether the identified shocks mirror the evidence in the oil market. Regarding oil supply shocks, there are huge positive spikes in the 2004m10 and 2004m11 and these have been attributed to reduction in production by non-Organization of Petroleum Exporting countries (OPEC) as well as the uncertainty about crude oil supplies from volatile political violence and geopolitical risks in exporting countries in the case of Nigeria, Iraq and Venezuela. In addition, the positive supply shocks in the early parts of 2010 could be driven by the Arab Spring (British Petroleum, 2016). This was followed by huge negative shocks after the Arab Spring. Further, the identified negative supply shocks in 2015 are a reflective of crude oil supply exceeding global demand. The market recorded its highest global inventory in every quarter in 2015, with surplus accumulation of 1.72 million barrels per day (Energy Information Administration, 2015). Similarly, the aggregate demand shocks plummeted in the early and latter parts of 2010 and this could be driven by the world economic recession. The estimated shocks particularly in 2008 imply that plummeting price of crude oil emanates partly from adverse aggregate demand shocks and predominantly from oil market-specific negative shocks. This confirms Hamilton (2009) conclusion that price of crude oil exceeded the value influenced by market demand and supply conditions when it increased above US$ 140 in June 2008.

4.2. Impulse responses

We show the structural evidence on macroeconomic effects of the identified shocks on Ghana's macroeconomic aggregates. The SVAR model makes use of the identified oil shocks on five macroeconomic variables (real composite index for economic activity, inflation, real effective exchange rate, and money supply and interest rate). The oil shocks were identified on the assumption that unanticipated instability of oil price is caused by exogenous factors compared to seasonal variations of the rest of the factors in the model. **Figures 2–4** display the computed cumulative impulse response functions (IRFs) of the macroeconomic aggregates to the identified oil market shocks. The IRFs trace the impact of the identified exogenous shocks on the chosen macroeconomic indicators over time and are inferred to be the impact of the observed shocks on the macroeconomic variables. Following the wild bootstrap technique proposed by

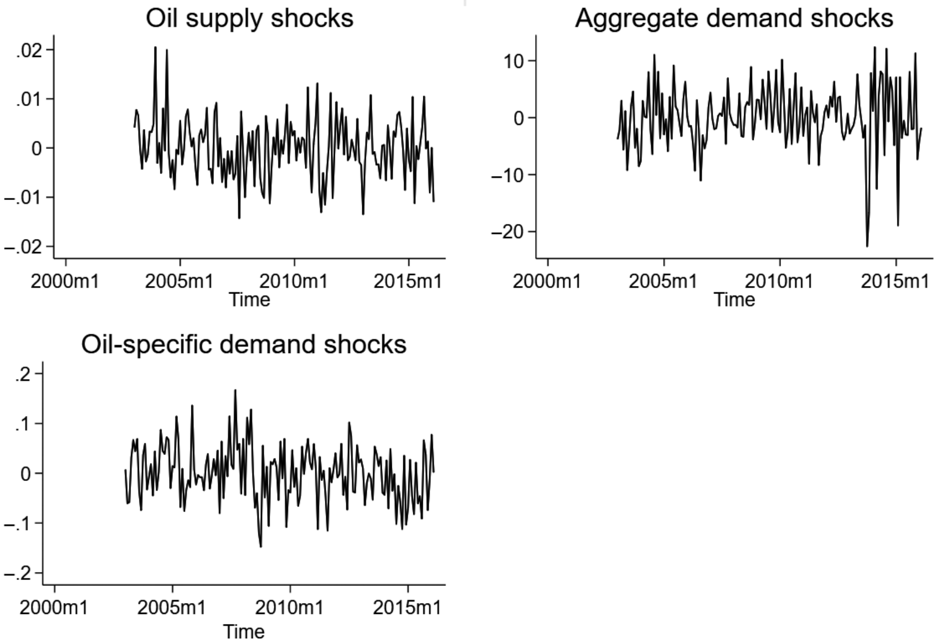

Figure 1 Sources of oil market shocks.
Source: Authors' own.

Gonçlves and Kilian (2004), we use one standard deviation confidence intervals on the vertical axis and number of months on the horizontal axis. The magnitude of the identified shocks is normalised in order that a structural innovation represents 1 per cent effect of an increase or decrease of oil price on impact. The dotted lines relate to plus or minus standard errors round the impulse responses, while the black lines denote the responses.

The effect is remarkably dissimilar for the various kind of shocks. For instance, oil supply shock possesses a weighty effect on GDP. A 1 per cent unfavourable oil supply shock leads to a temporary fall in GDP growth by about 0.024 per cent in the first month which is followed by a reversal by the close of the year. This implies structural innovation has contractionary effect and renders economic activity unstable temporary. Similarly, an unexpected oil supply shock leads to an appreciation of the real effective exchange rate in the first month but reverses progressively in the second to sixth months before rising again. This can be attributed to the small nature of Ghana's economy and the importation of both crude and refined oil. In addition, a 1 per cent unfavourable oil

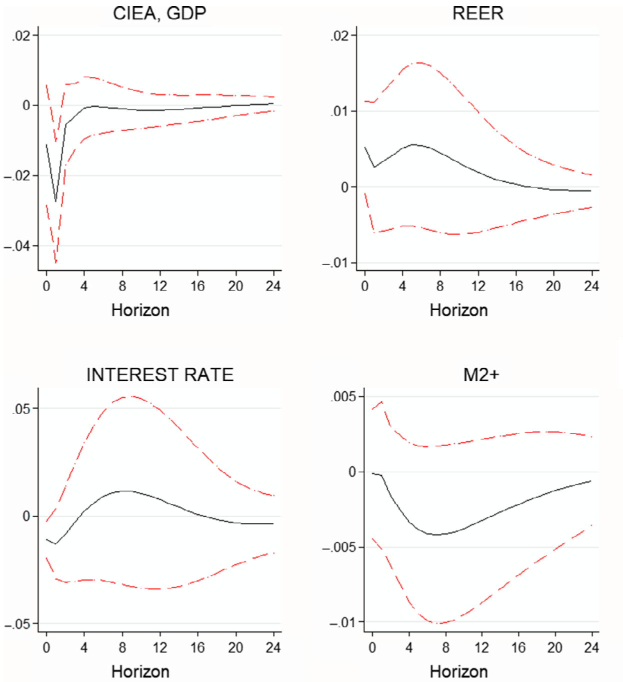

Figure 2 Reactions of macroeconomic indicators to oil supply disturbances. Note: Graphical display indicates of aggregated responses computed from equations (3) to (7). [Colour figure can be viewed at wileyonlinelibrary.com]

supply triggers a 0.025 per cent rise in the level of consumer price in the first month. This implies a trade-off between output stabilisation and low level of inflation in the short term. To reduce the harmful impact of inflation, monetary policy is tightened as the nominal interest rates rise to about 0.021 per cent in the ninth month. Finally, unexpected oil supply shock leads to M2+ shifting down on impact and continues to be negative for a protracted period of months.

In **Fig. 3**, a positive world total demand shock triggers a momentary rise of GDP, while effective exchange rate and general price level rise permanently. This outcome confirms Kilian (2009) who concluded that during the short run, the direct motivating impact of increased world demand on an economy will overshadow the indirect growth impeding consequences of increasing oil price. However, after some time, the impact decays and the negative effects of the higher oil price driven by this shock lead and cause the initial positive aggregate demand shock to generate a recessionary effect with a lag. Even though both the undeviating and indirect effects exhibit opposite sign on real GDP,

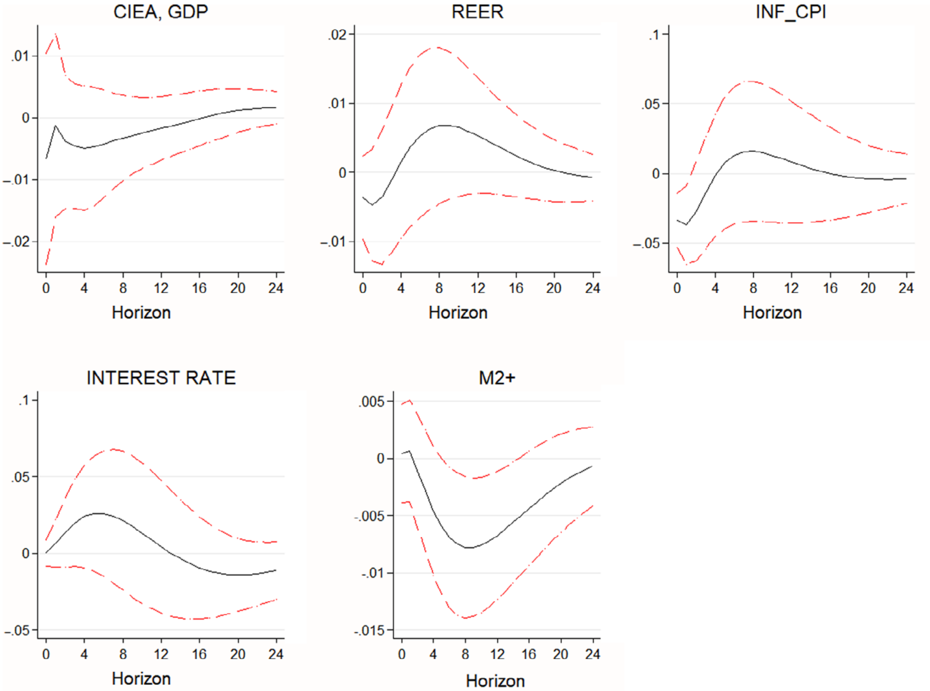

Figure 3 Reactions of macroeconomic indicators to total demand shocks. Note: Graphical display indicates aggregate responses computed from equations (3) to (7). [Colour figure can be viewed at wileyonlinelibrary.com]

their cumulative effect increases the price level. Similarly, the interest rate responds positively with a favourable aggregate demand shock. This signifies the tightening of monetary policy. The result of the contractionary monetary policy is a decrease in real money balances as the M2 + reduces continuously.

Furthermore, the macroeconomic effect of oil-specific demand disturbances generates outcomes that are not identical to the identified first two shocks. This shock causes a weighty decrease GDP particularly in the long term. Moreover, it triggers an appreciation of the exchange rate in the first two months of the year. This finding confirms Peersman and Van Robays (2009) who suggested that such an appreciation is driven by the propensity to invest in asset commodities as a safe haven against possible depreciation of the exchange rate. The appreciation of the exchange rate contributes to about 0.04 per cent transmission to consumer price level in the second month. However, a protracted depreciation of the exchange rate beginning from the third month transmits

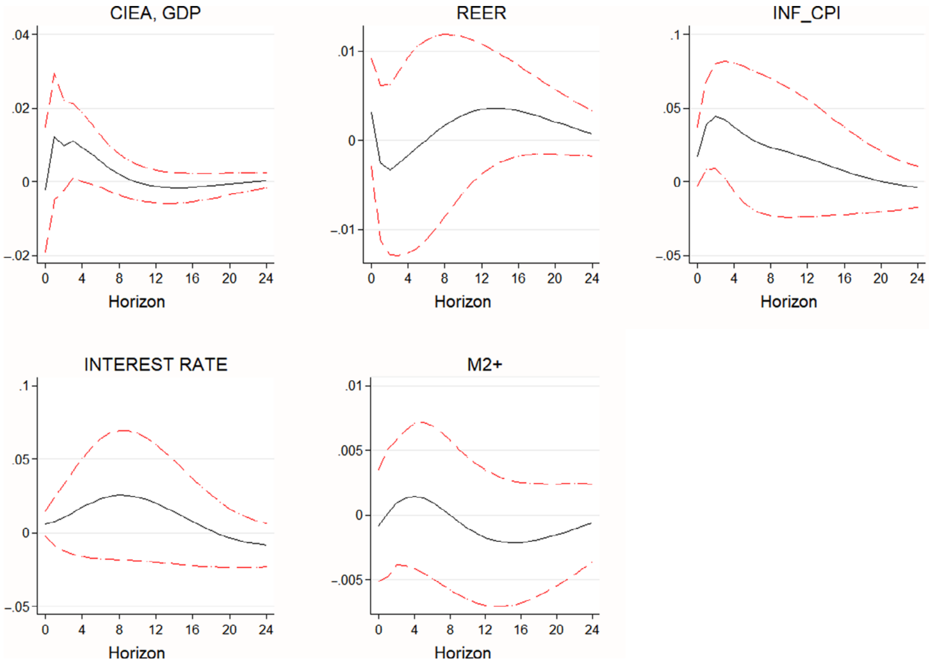

Figure 4 Reactions of macroeconomic indicators to oil-specific demand shocks. Note: Graphical display indicates aggregated responses computed from equations (3) to (7). [Colour figure can be viewed at wileyonlinelibrary.com]

to about a negligible of 0.012 per cent increases in consumer prices but a significant transmission to output by the close of the year. A depreciation of the real effective exchange rate then comoves with a high demand for oil and that is precisely a reflection of the behaviour of the oil-specific demand shock. Even though the effect on inflation is not significant, the reduction in output is significant.

Thus far, the evidence provided by **Figs 2–4** assists in understanding the fact that responses of macroeconomic aggregates in Ghana to the identified crude oil market shocks have proven to be unstable. This implies that the uncertainty of the estimates is copiously higher, which is reflective of the wider bands. Since it is the structure of demand and supply disturbances which cause oil price to rise over time, and that each disturbance is driven by a different response, we would expect the macroeconomic aggregates to be unstable. This is measured and shown by the gaps between the dotted lines and the response function.

4.3. Crude oil shocks and exchange rates

This section reports results about how several bilateral exchange rates in Ghana react in different ways to oil market shocks. This is important since oil is usually priced in USD, and therefore, a net oil consumer or producer may be motivated to increase the US dollar price when it loses value against other global major trading currencies. In addition, higher oil price entails a huge reallocation of wealth between an importing and exporting country (transfer of wealth from a consuming country to a producing country). Most importantly, it causes current surplus to the net exporting country but a current account deficit to net importing country. This has the potential of depreciating the currency of the importing country in the long term due to the adverse trade balance (Basher *et al.*, 2012). In examining this hypothesis, we compute the cumulative impulse response functions for the GHS/USD exchange rate and other bilateral exchange rates up to 12 months ahead. Following Atems *et al.* (2015), we calculate the impulse response functions for the various bilateral exchange rates using the following model:

$$\mathrm{ER}_t = \theta_0 + \sum_{i=0}^{12} \theta_{1i}\varepsilon_{s,t-i} + \sum_{i=0}^{12} \theta_{2i}\varepsilon_{\gamma,t-i} + \sum_{i=0}^{12} \theta_{3i}\varepsilon_{\delta,t-i} + \mu_t \tag{9}$$

where ER_t represents a per cent variation in the exchange rate series, while $\varepsilon_{s,t}, \varepsilon_{\gamma,t}$ and $\varepsilon_{\delta,t}$ capture the identified oil supply, total demand and oil-specific demand shocks. Equation (8) also captures the 95 per cent confidence intervals of the series. This estimation technique entails a recovery of the structural shocks from the VAR and then estimating the collective impulse response functions using a dynamic ordinary least-squares methodology. According to Kilian (2009), this procedure gives a parsimonious effective identification strategy.

Figure 5 reports the reaction of the bilateral nominal exchange rates series to the identified structural shocks estimated from equation (8). The estimated results entail thought-provoking discussions. **Figure 5** indicates oil supply shock displays no important impact on GHS/USD and GHS/GBP, while it significantly drives the GHS/EURO nominal exchange rate particularly in the 12 months. Similarly, the aggregate demand and oil-specific demand shocks display significant effects on the GHS/EURO nominal exchange rate which implies oil market shocks drive the depreciation of the exchange rate. This evidence confirms Golub (1983) theoretical stock/flow wealth transfer model. This model states that a reallocation of wealth caused by an oil price shock which results in an increase in the use of an external currency and a fall in the use of domestic currency will drive the domestic currency to fall in value against the foreign currency. Contrarily, the findings showed no proof of a significant impact of oil market shocks on the GHS/USD and GHS/GBP exchange rates.

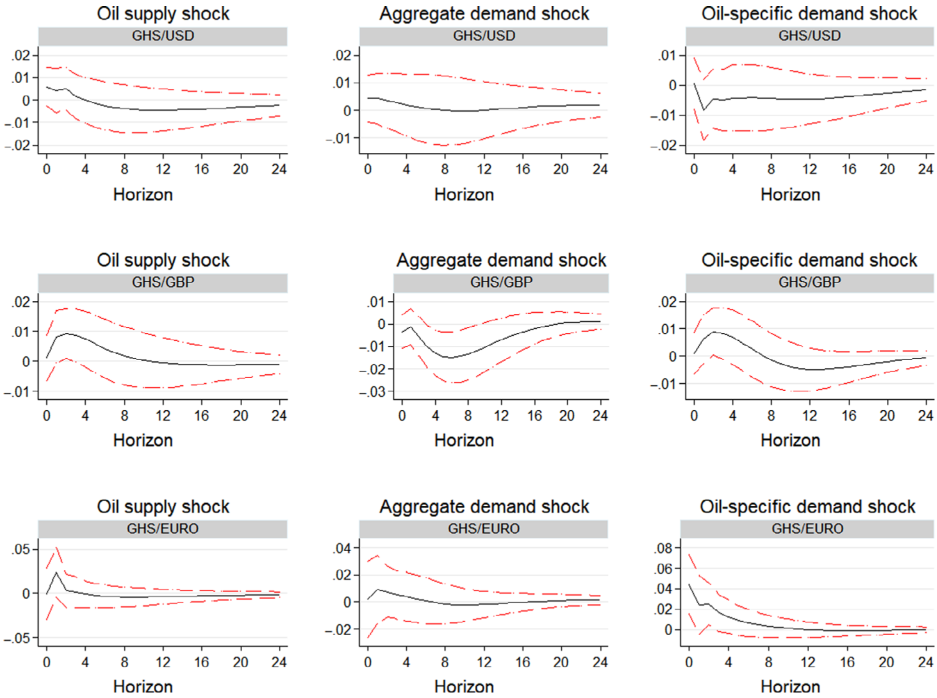

Figure 5 Reactions of bilateral exchange rate to each structural shock. [Colour figure can be viewed at wileyonlinelibrary.com]

4.4. Variance decomposition

Impulses response functions (IRFs) only provide information on the transmission of one-time oil market shocks but fail to show the effects of these shocks on the average. In addition, IRFs are unable to provide information on the extent of historical changes in oil market and macroeconomic aggregates that can be attributed to oil market shocks. To address these shortcomings of IRFs, we estimate the forecast error variance decomposition and past breakdown of the shocks using the SVAR model (see **Table 2**). The variance decomposition evaluates the role played by the shocks to the unpredictability of the macroeconomic aggregates. The results show that the magnitude of variance of change depends on the macroeconomic aggregates. For example, with respect to real composite index of economic activity supply shock explains much of the variance of change in the macroeconomic aggregates in Ghana followed by the oil-specific demand shock. Aggregate demand shock accounts for the least of the variance of change in the macroeconomic aggregates. For instance, within the 9th to the 24th month 5.6 per cent of

Table 2 Variance decomposition

Horizon	Fraction of CIEA variance due to			Fraction of REER variance due to			Fraction of INF variance due to			Fraction of IR variance due to			Fraction of M2+ variance due to		
	OSS	ODS	OSDS	OSS	ODS	OSDS	OSS	ODS	OSDS	OSS	ODS	OSDS	OSS	ODS	OSDS
1	0.0093	0.0031	0.0003	0.0155	0.0077	0.0057	0.0253	0.0582	0.0153	0.034	0.0582	0.0102	0.00002	0.0002	0.0007
3	0.0588	0.0038	0.0161	0.0087	0.0095	0.0053	0.0292	0.046	0.0534	0.014	0.046	0.0078	0.0017	0.0111	0.001
5	0.0578	0.0065	0.0289	0.0121	0.0071	0.005	0.0203	0.0305	0.0618	0.0063	0.0203	0.0117	0.0097	0.0144	0.0025
7	0.0569	0.0089	0.0337	0.0173	0.0109	0.0046	0.0158	0.0252	0.0607	0.0054	0.0278	0.0179	0.0205	0.0444	0.0031
9	0.0563	0.0103	0.0343	0.0209	0.0192	0.0044	0.0136	0.0246	0.0593	0.0064	0.0306	0.0249	0.03	0.0792	0.0289
11	0.0559	0.011	0.0339	0.0227	0.0277	0.0056	0.0124	0.0246	0.0589	0.0074	0.0301	0.0309	0.0368	0.1088	0.0304
13	0.0555	0.0113	0.0337	0.0233	0.0336	0.0079	0.0118	0.0243	0.059	0.0078	0.0282	0.035	0.0409	0.1292	0.043
17	0.055	0.0113	0.0338	0.0233	0.0381	0.0127	0.0114	0.0237	0.0592	0.0078	0.0237	0.0371	0.0439	0.1458	0.0849
24	0.0547	0.0118	0.0339	0.0232	0.0384	0.0158	0.0113	0.0238	0.0589	0.0113	0.0238	0.0377	0.0439	0.1457	0.012

Source: OSS, ODS and OSDS denote oil supply shock, oil demand shock and oil-specific demand shock.

the variance of change in real composite index of economic activity attributable to oil production shock, 3.3 per cent attributed to oil-specific demand shock, and 1.1 per cent is explained by aggregate demand shock and 90 per cent attributable to other shocks. Regarding exchange rate, total demand shock accounts for about 3.8 per cent of the variance of change, and supply shock contributes about 2.3 per cent, while market-specific shock contributes about 1.3 within the 13th to the 24th months.

4.5. Transmission effects of oil market shocks to food and non-food inflation

Next we evaluate the exact pass-through of oil markets shocks on food and non-food general price levels and deduce whether the pass-through are driven by direct or second-hand effects. We address this by decomposing the variance of the relative price to examine the degree to which variations in relative price indexes can be attributed to the identified oil market shocks. Following this procedure, the associated pass-through elasticity is estimated through a division of the aggregated impulse responses from food and non-food inflation by the preliminary oil market shocks. This is written as:

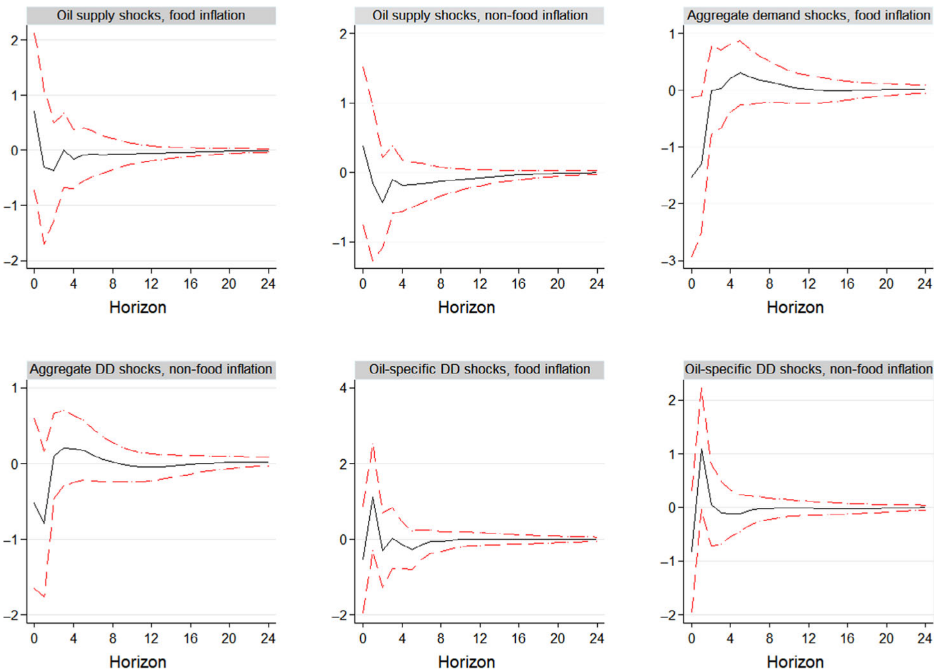

Figure 6 Responses of food and non-food price levels to each structural shock. [Colour figure can be viewed at wileyonlinelibrary.com]

$$\mathrm{PT}_t = \frac{\%\Delta P_t}{\%\Delta OMS_t} \tag{10}$$

where $\%\Delta P_{\mathrm{t}}$ denotes the percentage change in food and non-food inflation from period 0 to t in reaction to a preliminary oil market shocks, while $\%\Delta \mathrm{OMS}_t$ represents the percentage in the initial oil market shocks.

Figure 6 reveals a significant effect the structural shocks have on food and non-food inflation. For instance, a 1 per cent oil supply shock reduces food inflation by 0.2 per cent and non-food inflation by 0.31 per cent which later rise to about 0.1 per cent by the close of two years. Contrarily, a 1 per cent negative aggregate demand shock increases food and non-food inflation on impact by 0.23 per cent which later falls to about 0.1 per cent by the close of two years. Similarly, 1 per cent unfavourable oil-specific demand shock potentially increases food and non-food inflation on impact to about 1.2 per cent but fall to about 0.01 per cent by the end of two years. Since the transmission impacts of oil shocks to food and non-food inflation are significant, it implies inflation in Ghana is partly caused by direct effect of increasing oil prices in the consumer basket or indirect through the rising cost of production on non-oil goods and service. In addition, it means inflation targeting to a large extent has been successful in stabilising price levels in Ghana, particularly in the long term since the bands are narrower.

4.6. Discussion on the transmission mechanism

This section provides a brief discussion on the mechanism through which oil price shocks affect the macroeconomic variables presented in the preceding section. First, with respect to the effect on GDP, an unfavourable oil price shock leads to a fall in demand for oil as an input in the production of goods and services. This dampens economic activity leading to a fall in GDP since oil is an important input in the most productive sectors of the economy. Second, an unfavourable oil shock reduces disposable income which dampens economic activity leading to a fall in GDP. Third, the effect of oil price shocks on GDP is transmitted through the changes in the policy rate. The oil price shocks generated high inflation which led to initial tightening of monetary policy but later loosened to induce economic activity.

With respect to inflation, the high cost of oil increased production cost and exerted significant pressure on profits of firms. Price setting behaviour of firms served as an incentive for firms to raise their markup putting pressure on the general price level. Again, a lot of goods imported into Ghana are manufactured using oil as an input. Thus, an unfavourable oil price shock affected prices of these goods before importation into Ghana. This affected consumer prices and fuelled inflation in Ghana.

Table 3 Testing for asymmetries

A: Positive and negative shocks			
Macro aggregate	$\sum\lambda_j^+$	$\sum\theta_j^-$	$H_0:\sum\lambda_j=\sum\theta_j$
GDP (CIEA)	0.116	−0.02	5.85 [0.0167]
Real Effective Exchange rate	−0.016	0.006	1.13 [0.2886]
Inflation	−0.013	0.005	0.13 [0.7199]
Interest rate	−0.005	−0.006	0.00 [0.9657]
M2+	0.01	−0.006	0.88 [0.3503]

Source: Values in square parentheses are *P*-values.

Exchange rate movement mechanism is induced by import prices of oil which caused the import bill to rise relative to export bill. This then required the use of more or less local currency to be traded for foreign currency to enable Ghanaians to import oil and its related products leading to a depreciation or appreciation of the Ghana cedi.

The movement in money supply can be explained through domestic borrowing when the monetary is tightened or loosened. When Bank of Ghana tightened monetary policy due to high inflation, it raised policy rate which reduced the ability of consumers and firms to borrow funds which then reduced the supply of money. The supply of money later increased when monetary policy was relaxed and interest allowed to drop.

4.7. Threshold effects

We conclude our analyses by estimating the two threshold models presented in Sections 3.3 to test for asymmetries between favourable and adverse oil shocks on macroeconomic aggregates. Thus, for each macroeconomic aggregate examined in this study, we run equation (8) and test the following hypotheses: $\sum\lambda_j=\sum\theta_j$ and $\sum\alpha_j=\sum\alpha_j$. Generally, the results presented in **Table 3** show absence of asymmetry for all the variables with the exception of real GDP. The size of the estimated impact of oil price increases or decreases on the macroeconomic aggregates is about the same, lending little support for such asymmetries to be present.

5. Conclusion and policy implications

The paper applied a two-stage technique suggested by Kilian (2009) to disaggregate oil market shocks into oil production shocks, total demand shocks and oil-specific demand shocks and examined how macroeconomic aggregates responded to these identified shocks. Further, we examined these identified shocks on bilateral exchange rates such as

Ghs/USD, Ghs/GBP and Ghs/Euro to find out whether Ghana may be motivated to increase the GH/US dollar price when the US dollar loses value relative to other global major trading currencies. The evidence revealed that the responses of the macroeconomic aggregates are not similar because of the different underlying shocks of oil price movements. Generally, the study indicated that oil supply shock generated significant unfavourable effect on GDP, implying that real GDP plummeted following oil supply shock. Similarly, oil-specific demand shock possessed a significant harmful effect on real GDP. But, the total demand shock does not generate significant impact on the macroeconomic aggregates of Ghana. Further, the analysis revealed that the identified oil shocks generated significant effect on the GHs/Euro which indicated that the GHs/Euro exchange rate depreciates on effect, in the presence of crude oil market shocks. Generalised forecast variance decomposition evaluation confirmed the outcomes produced by IRFs. The results revealed that oil price shocks affect interest rate, inflation, output and exchange rate. Nevertheless, GDP dominated in variations. This means that external shocks emanating from demand and supply sides are important drivers of economic stagnation in Ghana. More importantly, the uncertainty of the estimates is copiously higher, which is a reflection of the wider bands and therefore suggested that crude oil market shocks rendered most macroeconomic aggregates unstable.

We also investigated the pass-through of the identified structural shocks on food and non-food price levels. The evidence suggested that all the identified structural shocks have significant impacts on the disaggregated inflation in Ghana. Therefore, direct effect of increasing oil prices in the consumer basket or indirect through the rising cost of production on oil intensive and non-oil intensive goods and service accounted for inflation in Ghana. This implied that oil market shocks are detrimental to food and non-food inflation in Ghana but inflation targeting adopted by the monetary authorities has moderated these impacts to ensure price stability.

Notes

1. The capital development cost and daily operating and technical costs of Jubilee Fields for 26 years are about US$10 billion. Ghana is contributing US$2 billion of this amount in 15 years.
2. The precautionary demand is derived via structural decomposition of the real oil prices. Upon extracting the supply and aggregate demand components of changes to the real price of oil, the residuals are termed precautionary demand shock.
3. The Ghana Statistical Service started publishing quarterly GDP in 2010 which is inadequate to perform any meaningful time series estimations. In addition, the study is unable to use interpolated quarterly real GDP since it has the potential of generating spurious dynamics.

4. Kilian (2009) has provided justifications underlying the assumption that the structural shocks are treated as predetermined.
5. Inflation is decomposed into food and non-inflation.
6. The study also estimates the effect of structural oil shock on three bilateral nominal exchange rates such as the Ghana Cedis/U.S. Dollar, Ghana Cedis/British Pound Sterling and Ghana Cedis/Euro.
7. Data on global economic activity were sourced from Kilian website: http://www-personal.umich.edu/-lkilain/reaupdate.txt.

References

British Petroleum. 2016. *Annual Report*. BP, London.

Energy Commission, 2016. 'Annual report'.

Ministry of Finance, 2017. 'Petroleum receipts and distribution report for 2nd quarter of 2017'.

Atems, B., Kapper, D. and Lam, E., 2015. Do exchange rates respond asymmetrically to shocks in the crude oil market? *Energy Economics* 49, 227–238.

Balke, N.S., Brown, S.P. and Yuucel, M.K., 2002. Oil price shocks and the us economy: Where does the asymmetry originate? *The Energy Journal* 27–52.

Barsky, R.B. and Kilian, L., 1970. Oil and the macroeconomy since the 1970s. *Journal of Economic Perspectives* 18, 4, 115–134.

Basher, S.A., Haug, A.A. and Sadorsky, P., 2012. Oil prices, exchange rates and emerging stock markets. *Energy Economics* 34, 1, 227–240.

Basnet, H.C. and Upadhyaya, K.P., 2015. Impact of oil price shocks on output, inflation and the real exchange rate: evidence from selected asean countries. *Applied Economics* 47, 29, 3078–3091.

Baumeister, C., Peersman, G., Van Robays, I., 2010. The economic consequences of oil shocks: differences across countries and time. In Fry, R., Jones, C. and Kent, C. (eds) *Inflation in an Era of Relative Price Shocks*, Reserve Bank of Australia, pp. 91–128.

Bernanke, B.S., 1983. Irreversibility, uncertainty, and cyclical investment. *The Quarterly Journal of Economics* 98, 1, 85–106.

Bernanke, B., Gertler, M. and Watson, M.W., 2004. Oil shocks and aggregate macroeconomic behavior: The role of monetary policy: A reply. *Journal of Money, Credit, and Banking* 36, 2, 287–291.

Bernanke, B.S., Gertler, M., Watson, M., Sims, C.A. and Friedman, B.M., 1997. Systematic monetary policy and the effects of oil price shocks. *Brookings Papers on Economic Activity* 1, 91–157.

Blanchard, O.J. and Gali, J., 2007. The macroeconomic effects of oil shocks: Why are the 2000s so different from the 1970s?Technical report, National Bureau of Economic Research.

Blanchard, O.J. and Riggi, M., 2013. Why are the 2000s so different from the 1970s? A structural interpretation of changes in the macroeconomic effects of oil prices. *Journal of the European Economic Association* 11, 5, 1032–1052.

Chen, S.-S. and Chen, H.-C., 2007. Oil prices and real exchange rates. *Energy Economics* 29, 3, 390–404.

Cunado, J. and De Gracia, F.P., 2005. Oil prices, economic activity and inflation: evidence for some Asian countries. *The Quarterly Review of Economics and Finance* 45, 1, 65–83.

Du, L., Yanan, H. and Wei, C., 2010. The relationship between oil price shocks and china's macro-economy: An empirical analysis. *Energy Policy* 38, 8, 4142–4151.

Engemann, K.M., Owyang, M.T. and Wall, H.J., 2014. Where is an oil shock? *Journal of Regional Science* 54, 2, 169–185.

Golub, S.S., 1983. Oil prices and exchange rates. *The Economic Journal* 93, 371, 576–593.

Gonçalves, S. and Kilian, L., 2004. Bootstrapping autoregressions with conditional heteroskedasticity of unknown form. *Journal of Econometrics* 123, 1, 89–120.

Granger, C. and Newbold, P., 1974. Spurious regression in econometrics. *Journal of Econometrics*.

Hamilton, J.D., 1983. Oil and the macroeconomy since World War II. *Journal of Political Economy* 91, 2, 228–248.

Hamilton, J.D., 1996. This is what happened to the oil price-macroeconomy relationship. *Journal of Monetary Economics* 38, 2, 215–220.

Hamilton, J.D., 2003. What is an oil shock? *Journal of Econometrics* 113, 2, 363–398.

Hamilton, J.D., 2009. Causes and consequences of the oil shock of 2007–08, Technical report, National Bureau of Economic Research.

Herrera, A.M. and Pesavento, E., 2009. Oil price shocks, systematic monetary policy, and the "great moderation". *Macroeconomic Dynamics* 13, 1, 107–137.

Jumah, A. and Pastuszyn, G., 2007. Oil price shocks, monetary policy and aggregate demand in ghana.

Kilian, L., 2009. Not all oil price shocks are alike: Disentangling demand and supply shocks in the crude oil market. *American Economic Review* 99, 3, 1053–69.

Leduc, S. and Sill, K., 2004. A quantitative analysis of oil-price shocks, systematic monetary policy, and economic downturns. *Journal of Monetary Economics* 51, 4, 781–808.

Lee, K., Ni, S. and Ratti, R.A., 1995. Oil shocks and the macroeconomy: the role of price variability. *The Energy Journal* 39–56.

Lorusso, M., Pieroni, L., 2018. Causes and consequences of oil price shocks on the UK economy. *Economic Modelling* 7272, 223–236.

Luütkepohl, H. 2005. *New introduction to multiple time series analysis*. Springer Science & Business Media.

Mork, K.A., 1989. Oil and the macroeconomy when prices go up and down: an extension of Hamilton's results. *Journal of political Economy* 97, 3, 740–744.

Nusair, S.A., 2016. The effects of oil price shocks on the economies of the gulf co-operation council countries: Nonlinear analysis. *Energy Policy* 91, 256–267.

Peersman, G. and Van Robays, I., 2009. Oil and the euro area economy. *Economic Policy* 24, 60, 603–651.

Rotemberg, J.J. and Woodford, M., 1996. Imperfect competition and the effects of energy price increases on economic activity, Technical report, National Bureau of Economic Research.

Sims, C.A., 1980. Macroeconomics and reality. *Econometrica: Journal of the Econometric Society* 1–48.

Śmiech, S., Papież, M. and Dąbrowski, M. A., 2015. Does the euro area macroeconomy affect global commodity prices? Evidence from a SVAR approach. *International Review of Economics & Finance* 39, 485–503.

Wang, Y.S., 2013. Oil price effects on personal consumption expenditures. *Energy Economics* 36, 198–204.

Appendix A
Plots of the series

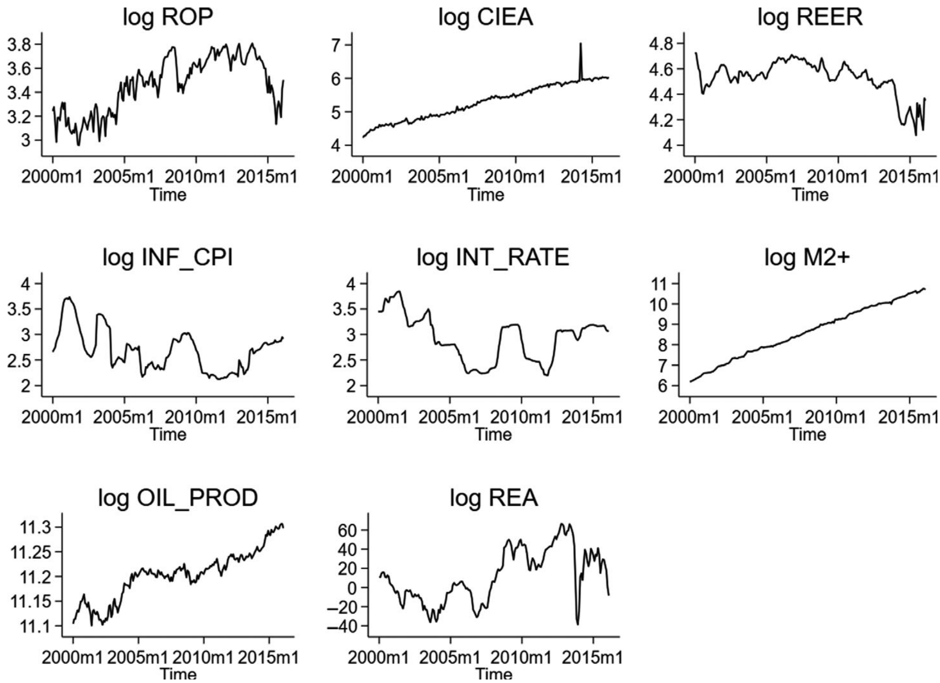

Figure A1 Plot of twelve seasonal differences of the variables. Source: Authors' own.

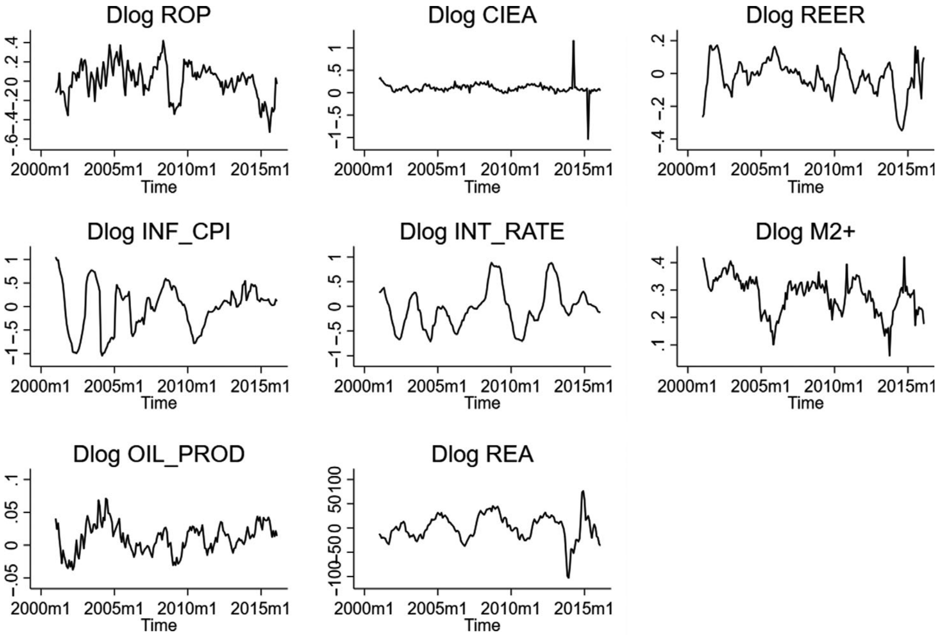

Figure A2 Plot of the variables in their levels. Source: Authors' own.

Appendix B
Unit root tests

Table B1 Augmented Dickey–Fuller unit roots test

	Level		Log first differences	
	Constant with trend	Constant with drift	Constant with trend	Constant with drift
lrop	−2.978	−2.286	−4.223	−3.992
lcieareal	−10.78	−1.92	−11.169	−11.05
loilprod	−2.908	−1.173	−3.609	−3.714
lrea	−1.961	−1.749	−4.484	−4.453
lreer	−2.349	−2.016	−4.256	−3.815
linf_cpi	−1.695	−1.61	−3.928	−3.752
lir_tb	−0.905	−1.286	−3.7	−3.511
lm2_plus	−3.072	−1.457	−2.941	−3.524
lghs/usd	−2.923	−1.104	−4.946	−4.091
lghs/gbp	−2.846	−1.475	−4.084	−3.887
lghs/euro	−9.066	−2.42	−6.647	−6.672
lfinf	−3.815	−2.15	−3.953	−4.026
lnfinf	−2.183	−2.222	−3.412	−3.891

Source: The Schwarz information criterion (SIC) was used to estimate optimal lag length for the tests. The 5 per cent critical value for the test with drift = −2.87 and trend = −3.42.